

System Requirements (SR.2)

Package in package 'Model'

System Requirements (SR.2)
Version 1.0 Phase 1.0 Proposed
peres created on 4/3/2026. Last modified 4/3/2026

System Requirements (SR.2) diagram

Requirements diagram in package 'System Requirements (SR.2)'

When a gesture or voice context is validated, the system shall send the corresponding home automation command to the central hub via Wi-Fi with confirmation of transmission.

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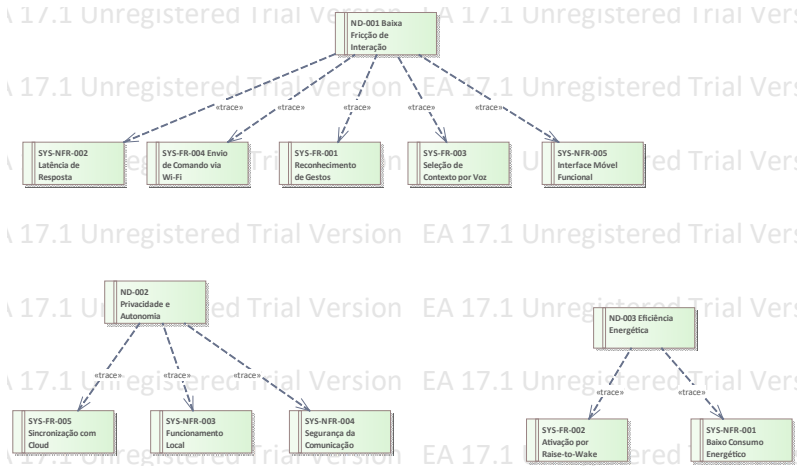


Figure 1: System Requirements (SR.2)

ND-001 Baixa Fricção de Interação

Requirement in package 'System Requirements (SR.2)'

O utilizador necessita de controlar dispositivos de automação residencial de forma rápida e intuitiva, sem depender exclusivamente de interações em ecrã ou de comandos de voz demorados.

ND-001 Baixa Fricção de Interação
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OUTGOING STRUCTURAL RELATIONSHIPS
Trace «trace» from ND-001 Baixa Fricção de Interação to SYS-NFR-005 Interface Móvel Funcional [Direction is 'Source -> Destination'.]

OUTGOING STRUCTURAL RELATIONSHIPS
Trace «trace» from ND-001 Baixa Fricção de Interação to SYS-FR-001 Reconhecimento de Gestos [Direction is 'Source -> Destination'.]
Trace «trace» from ND-001 Baixa Fricção de Interação to SYS-FR-003 Seleção de Contexto por Voz [Direction is 'Source -> Destination'.]
Trace «trace» from ND-001 Baixa Fricção de Interação to SYS-NFR-002 Latência de Resposta [Direction is 'Source -> Destination'.]
Trace «trace» from ND-001 Baixa Fricção de Interação to SYS-FR-004 Envio de Comando via Wi-Fi [Direction is 'Source -> Destination'.]

CONNECTORS
Trace «trace» Source -> Destination From: ND-001 Baixa Fricção de Interação : Requirement, Public To: SYS-NFR-005 Interface Móvel Funcional : Requirement, Public
Trace «trace» Source -> Destination From: ND-001 Baixa Fricção de Interação : Requirement, Public To: SYS-FR-001 Reconhecimento de Gestos : Requirement, Public
Trace «trace» Source -> Destination From: ND-001 Baixa Fricção de Interação : Requirement, Public To: SYS-FR-003 Seleção de Contexto por Voz : Requirement, Public
Trace «trace» Source -> Destination From: ND-001 Baixa Fricção de Interação : Requirement, Public To: SYS-NFR-002 Latência de Resposta : Requirement, Public
Trace «trace» Source -> Destination From: ND-001 Baixa Fricção de Interação : Requirement, Public To: SYS-FR-004 Envio de Comando via Wi-Fi : Requirement, Public

ND-002 Privacidade e Autonomia

Requirement in package 'System Requirements (SR.2)'

O utilizador necessita de um sistema capaz de processar a maior parte da interação localmente, reduzindo a dependência de serviços cloud, a latência de resposta e os riscos associados à privacidade.

ND-002 Privacidade e Autonomia

Version 1.0 Phase 1.0 Proposed

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OUTGOING STRUCTURAL RELATIONSHIPS

OUTGOING STRUCTURAL RELATIONSHIPS
Trace «trace» from ND-002 Privacidade e Autonomia to SYS-NFR-003 Funcionamento Local [Direction is 'Source -> Destination'.]
Trace «trace» from ND-002 Privacidade e Autonomia to SYS-NFR-004 Segurança da Comunicação [Direction is 'Source -> Destination'.]
Trace «trace» from ND-002 Privacidade e Autonomia to SYS-FR-005 Sincronização com Cloud [Direction is 'Source -> Destination'.]

CONNECTORS
Trace «trace» Source -> Destination From: ND-002 Privacidade e Autonomia : Requirement, Public To: SYS-NFR-003 Funcionamento Local : Requirement, Public
Trace «trace» Source -> Destination From: ND-002 Privacidade e Autonomia : Requirement, Public To: SYS-NFR-004 Segurança da Comunicação : Requirement, Public
Trace «trace» Source -> Destination From: ND-002 Privacidade e Autonomia : Requirement, Public To: SYS-FR-005 Sincronização com Cloud : Requirement, Public

ND-003 Eficiência Energética

Requirement in package 'System Requirements (SR.2)'

O utilizador necessita de um wearable de baixo consumo energético, capaz de operar diariamente com minimização de processamento contínuo e de captação de áudio desnecessária.

ND-003 Eficiência Energética

Version 1.0 Phase 1.0 Proposed

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OUTGOING STRUCTURAL RELATIONSHIPS
Trace «trace» from ND-003 Eficiência Energética to SYS-NFR-001 Baixo Consumo Energético [Direction is 'Source -> Destination'.]
Trace «trace» from ND-003 Eficiência Energética to SYS-FR-002 Ativação por Raise-to-Wake [Direction is 'Source -> Destination'.]

CONNECTORS
Trace «trace» Source -> Destination From: ND-003 Eficiência Energética : Requirement, Public To: SYS-NFR-001 Baixo Consumo Energético : Requirement, Public

CONNECTORS	
 Trace «trace»	Source -> Destination
From:	ND-003 Eficiência Energética : Requirement, Public
To:	SYS-FR-002 Ativação por Raise-to-Wake : Requirement, Public

SYS-FR-001 Reconhecimento de Gestos

Requirement in package 'System Requirements (SR.2)'

When the user performs a wrist gesture, the system shall classify the gesture locally and determine the corresponding command in less than 200 ms.

SYS-FR-001 Reconhecimento de Gestos

Version 1.0 Phase 1.0 Proposed

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INCOMING STRUCTURAL RELATIONSHIPS	
 Trace «trace» from ND-001 Baixa Fricção de Interação to SYS-FR-001 Reconhecimento de Gestos	[Direction is 'Source -> Destination'.]

CONNECTORS	
 Trace «trace»	Source -> Destination
From:	ND-001 Baixa Fricção de Interação : Requirement, Public
To:	SYS-FR-001 Reconhecimento de Gestos : Requirement, Public

SYS-FR-002 Ativação por Raise-to-Wake

Requirement in package 'System Requirements (SR.2)'

When the IMU detects a raise-to-wake motion, the system shall activate the microphone for contextual audio capture without continuous audio sampling.

SYS-FR-002 Ativação por Raise-to-Wake

Version 1.0 Phase 1.0 Proposed

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INCOMING STRUCTURAL RELATIONSHIPS	
 Trace «trace» from ND-003 Eficiência Energética to SYS-FR-002 Ativação por Raise-to-Wake	[Direction is 'Source -> Destination'.]

CONNECTORS	
 Trace «trace»	Source -> Destination
From:	ND-003 Eficiência Energética : Requirement, Public
To:	SYS-FR-002 Ativação por Raise-to-Wake : Requirement, Public

SYS-FR-003 Seleção de Contexto por Voz

Requirement in package 'System Requirements (SR.2)'

When the user utters a room or context keyword, the system shall identify the selected context and associate subsequent actions with that context.

SYS-FR-003 Seleção de Contexto por Voz

Version 1.0 Phase 1.0 Proposed

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INCOMING STRUCTURAL RELATIONSHIPS
 Trace «trace» from ND-001 Baixa Fricção de Interação to SYS-FR-003 Seleção de Contexto por Voz [Direction is 'Source -> Destination'.]
CONNECTORS
 Trace «trace» Source -> Destination From: ND-001 Baixa Fricção de Interação : Requirement, Public To: SYS-FR-003 Seleção de Contexto por Voz : Requirement, Public

SYS-FR-004 Envio de Comando via Wi-Fi

Requirement in package 'System Requirements (SR.2)'

When a gesture or voice context is validated, the system shall send the corresponding home automation command to the central hub via Wi-Fi with confirmation of transmission.

SYS-FR-004 Envio de Comando via Wi-Fi

Version 1.0 Phase 1.0 Proposed

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INCOMING STRUCTURAL RELATIONSHIPS
 Trace «trace» from ND-001 Baixa Fricção de Interação to SYS-FR-004 Envio de Comando via Wi-Fi [Direction is 'Source -> Destination'.]
CONNECTORS
 Trace «trace» Source -> Destination From: ND-001 Baixa Fricção de Interação : Requirement, Public To: SYS-FR-004 Envio de Comando via Wi-Fi : Requirement, Public

SYS-FR-005 Sincronização com Cloud

Requirement in package 'System Requirements (SR.2)'

When the wearable changes state, the system shall synchronize the new state with the cloud database to keep the dashboard updated.

SYS-FR-005 Sincronização com Cloud

Version 1.0 Phase 1.0 Proposed
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INCOMING STRUCTURAL RELATIONSHIPS

⇒ Trace «trace» from ND-002 Privacidade e Autonomia to SYS-FR-005 Sincronização com Cloud
[Direction is 'Source -> Destination'.]

CONNECTORS

Trace «trace» Source -> Destination
From: ND-002 Privacidade e Autonomia : Requirement, Public
To: SYS-FR-005 Sincronização com Cloud : Requirement, Public

SYS-NFR-001 Baixo Consumo Energético

Requirement in package 'System Requirements (SR.2)'

During normal operation, the wearable shall minimize continuous processing and audio capture to support daily use with low battery consumption.

SYS-NFR-001 Baixo Consumo Energético
Version 1.0 Phase 1.0 Proposed
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INCOMING STRUCTURAL RELATIONSHIPS

⇒ Trace «trace» from ND-003 Eficiência Energética to SYS-NFR-001 Baixo Consumo Energético
[Direction is 'Source -> Destination'.]

CONNECTORS

Trace «trace» Source -> Destination
From: ND-003 Eficiência Energética : Requirement, Public
To: SYS-NFR-001 Baixo Consumo Energético : Requirement, Public

SYS-NFR-002 Latência de Resposta

Requirement in package 'System Requirements (SR.2)'

After a valid command is recognized, the system shall trigger the corresponding action on the target device in less than 1 second.

SYS-NFR-002 Latência de Resposta
Version 1.0 Phase 1.0 Proposed
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INCOMING STRUCTURAL RELATIONSHIPS

⇒ Trace «trace» from ND-001 Baixa Fricção de Interação to SYS-NFR-002 Latência de Resposta
[Direction is 'Source -> Destination'.]

CONNECTORS	
 Trace «trace» Source -> Destination	
From:	ND-001 Baixa Fricção de Interação : Requirement, Public
To:	SYS-NFR-002 Latência de Resposta : Requirement, Public

SYS-NFR-003 Funcionamento Local

Requirement in package 'System Requirements (SR.2)'

The system shall perform the core gesture recognition locally on the wearable without requiring permanent external internet connectivity.

SYS-NFR-003 Funcionamento Local
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INCOMING STRUCTURAL RELATIONSHIPS	
 Trace «trace» from ND-002 Privacidade e Autonomia to SYS-NFR-003 Funcionamento Local	[Direction is 'Source -> Destination'.]

CONNECTORS	
 Trace «trace» Source -> Destination	
From:	ND-002 Privacidade e Autonomia : Requirement, Public
To:	SYS-NFR-003 Funcionamento Local : Requirement, Public

SYS-NFR-004 Segurança da Comunicação

Requirement in package 'System Requirements (SR.2)'

When data is transmitted between the wearable, the hub, and the dashboard, the system shall protect the communication against unauthorized access and data interception.

SYS-NFR-004 Segurança da Comunicação
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INCOMING STRUCTURAL RELATIONSHIPS	
 Trace «trace» from ND-002 Privacidade e Autonomia to SYS-NFR-004 Segurança da Comunicação	[Direction is 'Source -> Destination'.]

CONNECTORS	
 Trace «trace» Source -> Destination	
From:	ND-002 Privacidade e Autonomia : Requirement, Public
To:	SYS-NFR-004 Segurança da Comunicação : Requirement, Public

SYS-NFR-005 Interface Móvel Funcional

Requirement in package 'System Requirements (SR.2)'

When the user opens the mobile application, the system shall display the current device status and allow basic configuration and monitoring actions.

SYS-NFR-005 Interface Móvel Funcional
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INCOMING STRUCTURAL RELATIONSHIPS
 Trace «trace» from ND-001 Baixa Fricção de Interação to SYS-NFR-005 Interface Móvel Funcional [Direction is 'Source -> Destination'.]
CONNECTORS
 Trace «trace» Source -> Destination From: ND-001 Baixa Fricção de Interação : Requirement, Public To: SYS-NFR-005 Interface Móvel Funcional : Requirement, Public